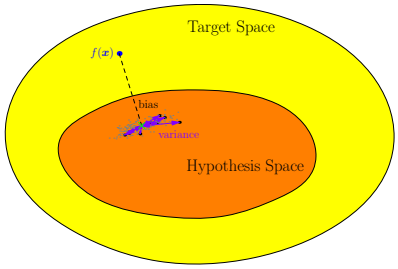
# Mathematics of Machine Learning

## *When Machine Learning Works*

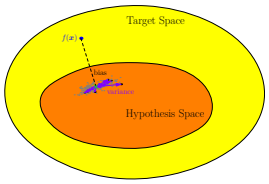



*When ML Works, Bias Variance*



# Outline

1. **What Makes a Good Learning Machine?**
2. Bias-Variance Dilemma
3. Shrinkage and Ridge Regression
4. Ensembles






# What Makes a Good Learning Machine?

- We want to understand why some machine learning techniques work well and others don't
- To understand why they work we need to understand what makes a good learning machine
- For this we have to get conceptual and think about **generalisation** performance

  ***generalisation**: how well do we do on unseen data as opposed to the training data*



# What Makes Machine Learning Hard?

- Typically we work in high dimensions (i.e. have many features)
- The problem can be over-constrained (i.e. we have conflicting data to deal with)—solve by minimising an error function
- The problem can be under-constrained (i.e. there are many possible solutions that are consistent with the data)—need to choose a plausible solution
- Typically in machine learning the data will be over-constrained in some dimensions and under-constrained in others
- We can't visualise the data to know what is going on



# Least Squares

- Suppose we want to learn some output $y$ for a feature vector $\boldsymbol{x}$
- We construct a learning machine that makes a prediction $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta})$
- We typically choose the machine to minimise a *training loss*

$$L_T(\mathcal{D}) = \sum_{(\boldsymbol{x},y)\in\mathcal{D}} \Big(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}) - y\Big)^2 = \sum_{i=1}^{m} \Big(\hat{f}(\boldsymbol{x}_i|\boldsymbol{\theta}) - y_i\Big)^2$$

  where $\mathcal{D} = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^m$ is a set of size $m$, sampled from a probability distribution $\mu(\boldsymbol{x},y)$

- We call this machine $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D})$

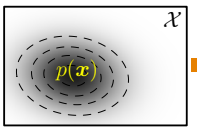


# Generalisation Error

- We want to minimise the *generalisation loss* which in this case is

$$L_G(\boldsymbol{\theta}_\mathcal{D}) = \sum_{(\boldsymbol{x},y)\in\mathcal{Z}} \mu(\boldsymbol{x},y)\Big(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - y\Big)^2$$

$$\mu(\boldsymbol{x},y) = p(y|\boldsymbol{x})p(\boldsymbol{x})$$



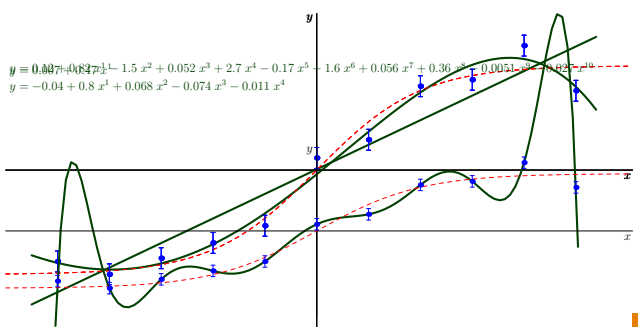

  (we can estimate this if we have some labelled examples $(\boldsymbol{x}_i, y_i)$ which we have not trained on)

- We want to minimise $L_G(\boldsymbol{\theta}_\mathcal{D})$ but in practice we areb minimising $L_T(\mathcal{D})$ *what could possibly go wrong?*



# Too Simple or Too Complex?

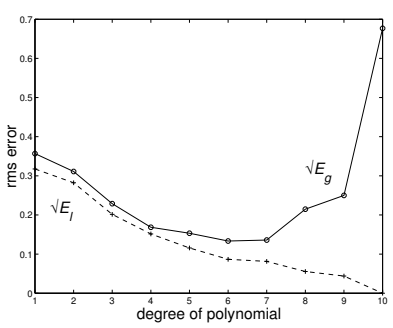
- Fit $\hat{f}(x|\boldsymbol{\theta}_\mathcal{D})$ to data





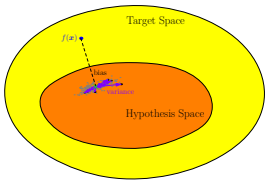

# Measuring Generalisation Error for Regression

- Consider the regression example. The root mean squared error is



## Outline


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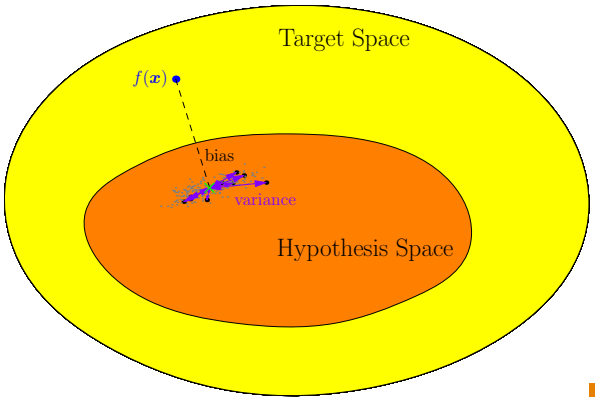



## Expected Generalisation Performance

- Our generalisation performance will depend on our training set, $\mathcal{D}$
- To reason about generalisation we can ask what is the *expected generalisation loss*, when we average over all different data sets of size $m$ drawn independently from $\mu(\boldsymbol{x},y)$
- For each data set, $\mathcal{D}$, we would learn a different approximator $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D})$
- Note that in practice we only get one data set. We might be lucky and do better than the expected generalisation or we might be unlucky and do worse



## Approximation and Estimation Errors






## Mean Machine

- To help understand generalisation we can consider the mean prediction with respect to machines trained with all data sets of size $m$

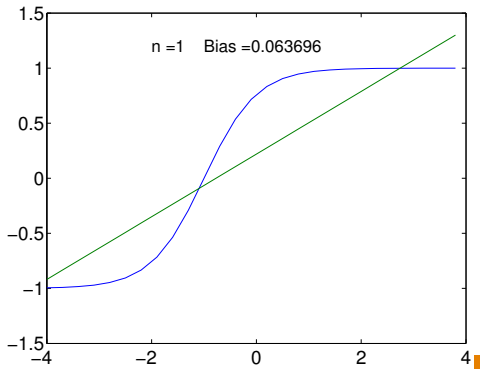
$$\hat{f}_m(\boldsymbol{x}) = \mathbb{E}_\mathcal{D}\Big[\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D})\Big]$$

- We can define the **bias** to be generalisation performance of the mean machine

$$B = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\Big(\hat{f}_m(\boldsymbol{x}) - y\Big)^2$$

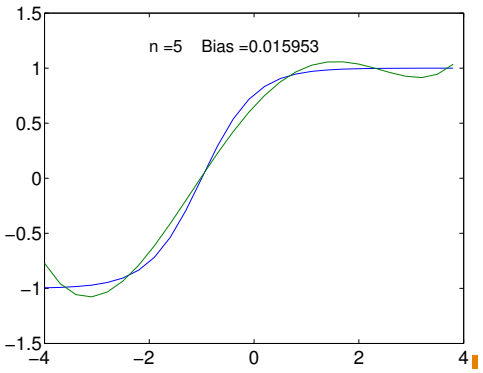


## Regression Example $n = 1$






## Regression Example $n = 5$






## Bias and Variance

Consider the expected generalisation for data sets of size $|\mathcal{D}| = m$

$$\begin{aligned}
\bar{L}_G = \mathbb{E}_\mathcal{D}[L_G(\boldsymbol{\theta}_\mathcal{D})] &= \mathbb{E}_\mathcal{D}\left[\sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\Big(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - y\Big)^2\right] \\
&= \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_\mathcal{D}\left[\Big(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - y\Big)^2\right] \\
&= \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_\mathcal{D}\left[\Big(\big(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - \hat{f}_m(\boldsymbol{x})\big) + \big(\hat{f}_m(\boldsymbol{x}) - y\big)\Big)^2\right] \\
&= \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\Bigg(\mathbb{E}_\mathcal{D}\left[\Big(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - \hat{f}_m(\boldsymbol{x})\Big)^2 + \Big(\hat{f}_m(\boldsymbol{x}) - y\Big)^2\right] \\
&\qquad + 2\mathbb{E}_\mathcal{D}\Big[\Big(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - \hat{f}_m(\boldsymbol{x})\Big)\Big(\hat{f}_m(\boldsymbol{x}) - y\Big)\Big]\Bigg)
\end{aligned}$$



## Cross Term

- The cross term vanishes

$$\begin{aligned}
C &= \mathbb{E}_\mathcal{D}\Big[\Big(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - \hat{f}_m(\boldsymbol{x})\Big)\Big(\hat{f}_m(\boldsymbol{x}) - y\Big)\Big] \\
&= \mathbb{E}_\mathcal{D}\Big[\Big(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - \hat{f}_m(\boldsymbol{x})\Big)\Big]\Big(\hat{f}_m(\boldsymbol{x}) - y\Big) \\
&= \Big(\mathbb{E}_\mathcal{D}\Big[\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D})\Big] - \hat{f}_m(\boldsymbol{x})\Big)\Big(\hat{f}_m(\boldsymbol{x}) - y\Big) \\
&= \Big(\hat{f}_m(\boldsymbol{x}) - \hat{f}_m(\boldsymbol{x})\Big)\Big(\hat{f}_m(\boldsymbol{x}) - y\Big) = 0
\end{aligned}$$

- Thus

$$\bar{L}_G = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_\mathcal{D}\left[\Big(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_\mathcal{D}) - \hat{f}_m(\boldsymbol{x})\Big)^2 + \Big(\hat{f}_m(\boldsymbol{x}) - y\Big)^2\right]$$

## Bias and Variance

- We can write the expected generalisation loss as

$$\mathbb{E}_{\mathcal{D}}[L_G(\boldsymbol{\theta}_{\mathcal{D}})] = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)^2\right] + \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\left(\hat{f}_m(\boldsymbol{x}) - y\right)^2 = V + B$$

- Where $B$ is the bias and $V$ is the variance defined by

$$V = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)^2\right]$$



## Where Is the Noise?

- Suppose $y = f^*(\boldsymbol{x}) + \epsilon$ where the noise $\epsilon$ has mean 0 and variance $\sigma^2$
- Then (the cross term vanishes again, as $\mathbb{E}[\epsilon] = 0$)

$$B = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\left(\hat{f}_m(\boldsymbol{x}) - y\right)^2 = \sigma^2 + \sum_{\boldsymbol{x}} \mu(\boldsymbol{x})\left(\hat{f}_m(\boldsymbol{x}) - f^*(\boldsymbol{x})\right)^2$$

- So $\bar{L}_G = \text{noise} + \text{bias}^2 + \text{variance}$, the form you may have seen elsewhere
- The noise is the same for every machine: we can do nothing about it
- Here we keep it inside $B$: $\bar{L}_G = B + V$



## Bias-Variance Dilemma

- The bias measures the generalisation performance of the *mean machine* and is large if the machine is too simple to capture the changes in the function we want to learn
- The variance measures the variation in the prediction of the machines as we change the data set we train on

$$V = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x},y)\mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)^2\right]$$

- The variance is usually large if we have a complex machine
- Striking the right balance is often the key to getting good results



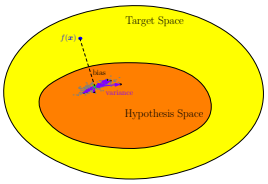
## Balancing Bias and Variance

- We want to choose a learning machine that is complex enough to capture the underlying function we are trying to learn, but otherwise as simple as possible
- There are a number of tricks to achieve this balance
- Some require us to preprocess the data to reduce the number of inputs
- Some machines cleverly adjust their own complexity
- The rest of this lecture: two exact examples, and averaging



## Outline

1. What Makes a Good Learning Machine?
2. Bias-Variance Dilemma
3. **Shrinkage and Ridge Regression**
4. Ensembles






## Shrinking an Estimate

- The simplest learning problem: estimate the mean $\theta$ of a distribution with variance $\sigma^2$ from $m$ samples
- The sample mean $\bar{x}$ is unbiased with variance $\sigma^2/m$
- Try the deliberately biased $\hat{\theta} = c\bar{x}$ with $0 \le c \le 1$

$$\mathbb{E}\left[\left(\hat{\theta} - \theta\right)^2\right] = \underbrace{(1-c)^2\theta^2}_{\text{bias}^2} + \underbrace{c^2\frac{\sigma^2}{m}}_{\text{variance}}$$

- At $c = 1$ the derivative is $2\sigma^2/m > 0$: shrinking a little **always** reduces the error
- The best $c$ depends on the unknown $\theta$, so in practice we choose it by validation



## Ridge Regression

- Data $\boldsymbol{y} = \mathbf{X}\boldsymbol{w}^* + \boldsymbol{\epsilon}$, with noise of variance $\sigma^2$ on each target
- Ridge regression minimises $\|\mathbf{X}\boldsymbol{w} - \boldsymbol{y}\|^2 + \eta\|\boldsymbol{w}\|^2$, giving

$$\hat{\boldsymbol{w}} = \left(\mathbf{X}^\mathsf{T}\mathbf{X} + \eta\mathbf{I}\right)^{-1}\mathbf{X}^\mathsf{T}\boldsymbol{y}$$

- $\eta = 0$ is least squares: unbiased, but how variable?
- Use the SVD $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^\mathsf{T}$ and work in the basis of the columns of $\mathbf{V}$, where everything separates



## Ridge Through the SVD

- Write $w_k^* = \boldsymbol{v}_k^\mathsf{T}\boldsymbol{w}^*$ for the component of $\boldsymbol{w}^*$ along $\boldsymbol{v}_k$; since $\boldsymbol{u}_k^\mathsf{T}\boldsymbol{y} = s_k w_k^* + \boldsymbol{u}_k^\mathsf{T}\boldsymbol{\epsilon}$, the same component of the estimate is

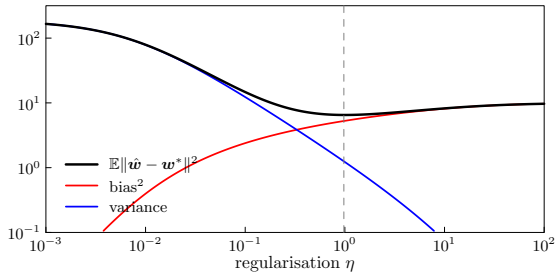
$$\hat{w}_k = \frac{s_k\boldsymbol{u}_k^\mathsf{T}\boldsymbol{y}}{s_k^2 + \eta}, \qquad \mathbb{E}[\hat{w}_k] = \frac{s_k^2}{s_k^2+\eta}w_k^*, \qquad \text{Var}(\hat{w}_k) = \frac{\sigma^2 s_k^2}{(s_k^2+\eta)^2}$$

- So the expected squared error of the weights is

$$\mathbb{E}\left[\|\hat{\boldsymbol{w}} - \boldsymbol{w}^*\|^2\right] = \sum_k \underbrace{\left(\frac{\eta}{s_k^2+\eta}\right)^2 w_k^{*2}}_{\text{bias}^2} + \underbrace{\frac{\sigma^2 s_k^2}{(s_k^2+\eta)^2}}_{\text{variance}}$$

- With $\eta = 0$ the variance is $\sum_k \sigma^2/s_k^2$: small singular values (ill-conditioning) make it huge! $\eta$ shrinks exactly those directions, trading a little bias for a large drop in variance

## Choosing $\eta$

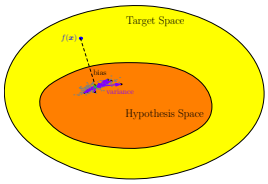



- Ten singular values from 3 down to 0.1, $\sigma = 1$: least squares has error 188, the best $\eta \approx 1$ gives 6.5
- In practice choose $\eta$ by cross-validation



## Outline

1. What Makes a Good Learning Machine?
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## Averaging Reduces Variance

- An **ensemble** averages $M$ predictors, $\bar{f}_M(\boldsymbol{x}) = \frac{1}{M}\sum_{i=1}^M f_i(\boldsymbol{x})$, trained with different randomness
- Averaging does not change the mean, so the bias is unchanged
- If each $f_i(\boldsymbol{x})$ has variance $s^2$ and each pair has correlation $\rho$

$$\mathrm{Var}\big(\bar{f}_M(\boldsymbol{x})\big) = \rho s^2 + \frac{1-\rho}{M}s^2$$

- More members remove the second term, but correlation sets a floor $\rho s^2$
- So we want members that are good **and** different



## Bagging, Forests and Deep Ensembles

- **Bagging**: train each member on a bootstrap resample of the data (sampled with replacement)
- **Random forests**: bagged decision trees that also choose each split from a random subset of the features—each tree is worse, but the trees are less correlated
- **Deep ensembles**: train several networks on the same data from different random initialisations
- The spread of the members' predictions is a useful measure of uncertainty
- Ensembles of high-variance, low-bias machines are among the most reliable methods in practice



## Lessons

- The expected generalisation loss splits into bias plus variance
- One of the most useful intuitions about what works is the bias-variance framework
- A biased estimate can beat an unbiased one: shrinkage, ridge regression
- The bias is high for simple machines that can't capture the data
- The variance is high for complex machines that are sensitive to the training set
- Averaging (ensembles) reduces variance without adding bias